Year 12 Further Maths - Roots of polynomials

Similar questions - Symmetric functions and related equations

Two new questions on this topic, a little more open-ended than the exam questions. They are not past exam questions. Answers are at the end.

1 **In this question you must show detailed reasoning.** You are given that α, β, γ are the roots of $3x^3 - x^2 + 2x - 4 = 0$.

(a) Find the value of $\alpha^2\beta^2 + \beta^2\gamma^2 + \gamma^2\alpha^2$. **[5]**

(b) Find a cubic equation whose roots are α^2, β^2 and γ^2, giving your answer in the form $ay^3 + by^2 + cy + d = 0$ with integer coefficients. **[4]**

2 **In this question you must show detailed reasoning.** The roots of $3x^2 - 5x + 9 = 0$ are α and β. By considering the symmetric functions $\alpha + \beta$ and $\alpha\beta$, determine the exact value of $\dfrac{1}{\alpha^2} + \dfrac{1}{\beta^2}$. **[4]**

3 **In this question you must show detailed reasoning.** The roots of $3x^3 - 2x^2 + x + 4 = 0$ are α, β and γ. By considering $\alpha^2 + \beta^2 + \gamma^2$, $\sum \alpha^{-2}$ and $\alpha\beta\gamma$, determine a cubic equation with integer coefficients whose roots are $\dfrac{\alpha\beta}{\gamma}$, $\dfrac{\beta\gamma}{\alpha}$ and $\dfrac{\gamma\alpha}{\beta}$. **[6]**

4 The equation $z^3 + pz^2 + qz + 30 = 0$, where p and q are real, has a root $1 + 3\text{i}$. Find p and q and the other roots. **[5]**

Question 1

(a) $\sum\alpha=\frac{1}{3}$, $\sum\alpha\beta=\frac{2}{3}$, $\alpha\beta\gamma=\frac{4}{3}$. $\sum\alpha^2\beta^2=(\sum\alpha\beta)^2-2\alpha\beta\gamma\sum\alpha=\frac{4}{9}-2\times\frac{4}{3}\times\frac{1}{3}=-\frac{4}{9}$

(b) $\sum\alpha^2=\left(\frac{1}{3}\right)^2-2\left(\frac{2}{3}\right)=-\frac{11}{9}$; $\sum\alpha^2\beta^2=-\frac{4}{9}$; $\alpha^2\beta^2\gamma^2=\frac{16}{9}$. $y^3+\frac{11}{9}y^2-\frac{4}{9}y-\frac{16}{9}=0$, i.e. $9y^3+11y^2-4y-16=0$

Question 2

$\alpha+\beta=\frac{5}{3}$, $\alpha\beta=3$. $\dfrac{1}{\alpha^2}+\dfrac{1}{\beta^2}=\dfrac{\alpha^2+\beta^2}{\alpha^2\beta^2}=\dfrac{(\alpha+\beta)^2-2\alpha\beta}{(\alpha\beta)^2}=\dfrac{\frac{25}{9}-6}{9}=-\dfrac{29}{81}$

Question 3

$\sum\alpha=\frac{2}{3}$, $\sum\alpha\beta=\frac{1}{3}$, $\alpha\beta\gamma=-\frac{4}{3}$. Write $\frac{\alpha\beta}{\gamma}=\dfrac{\alpha\beta\gamma}{\gamma^2}$. Sum of new roots $=\alpha\beta\gamma\sum\dfrac{1}{\alpha^2}$ where $\sum\dfrac{1}{\alpha^2}=\dfrac{(\sum\alpha\beta)^2-2\alpha\beta\gamma\sum\alpha}{(\alpha\beta\gamma)^2}=$ $\dfrac{\frac{1}{9}+\frac{16}{9}}{\frac{16}{9}}=\frac{17}{16}$, giving $-\frac{4}{3}\times\frac{17}{16}=-\frac{17}{12}$. Products in pairs: $\frac{\alpha\beta}{\gamma}\cdot\frac{\beta\gamma}{\alpha}=\beta^2$, so the sum is $\alpha^2+\beta^2+\gamma^2=\frac{4}{9}-\frac{2}{3}=-\frac{2}{9}$. Product of all three $=\alpha\beta\gamma=-\frac{4}{3}$. Equation: $y^3+\frac{17}{12}y^2-\frac{2}{9}y+\frac{4}{3}=0$, i.e. $36y^3+51y^2-8y+48=0$

Question 4

Roots 1 ± 3i and r. Product of roots $=-30$: $(1+9)r=-30$, $r=-3$. Sum of roots $=-p$: $2-3=-p$, $p=1$. Sum of products in pairs $=q$: $(1+9)+(-3)(2)=4$. So $p=1$, $q=4$ and the other roots are $1-3$i and -3